

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Silicon photonics optical gyroscope

ANELLO Photonics SIPHOG is the world's first silicon photonics optical gyroscope. The SIPHOG is a low-noise



and low-drift optical gyroscope smart sensor leveraging a photonic integrated circuit (IC) technology. This makes it the smallest optical gyroscope in the world, which enables reliable and accurate navigation and positioning across many autonomous applications. The SIPHOG can enable companies to design more stable, accurate, and robust sensors for various autonomous applications. It also features Anello's Sensor Fusion Engine, which delivers a full end-to-end offering to customers in various fields of autonomy where position, navigation, and timing are essential.

ANELLO Photonics

<https://www.anellophotonics.com>

Multi-function PMIC

Nordic Semiconductor's nPM1300 is a multi-function power management IC (PMIC) that supports both charging



of larger batteries and four regulated power rails. It comes in a small size measuring just 3.1mm × 2.4mm. The nPM1300 features hard reset, battery fuel gauge, system-level watchdog,

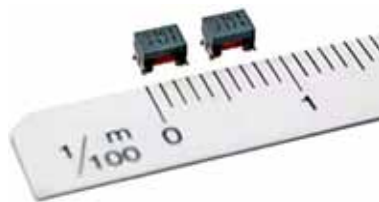
power loss warning, and recovery from failed boot. It is digitally configurable through an I2C-compatible two-wire interface (TWI). The PMIC also features port detection and automatic current limit up to 1500mA through USB-C power delivery. The nPM1300 PMIC simplifies the work of system designers by integrating longer run time and efficient battery charging and featuring essential functions required for Bluetooth Low Energy embedded designs.

Nordic Semiconductor

<https://www.nordicsemi.com>

Common mode automotive filter

TDK Corporation ACT1210E series is the industry's first common mode filter for automotive Ethernet 10Base-T1S. The ACT1210E series achieves the industry's best high S-parameters



(scattering parameters) and a maximum line-to-line capacitance of 10pF. The operating temperature range is from -40°C to +125°C. Laser welding winding wires to the metalising terminals provides high thermal shock resistance and outstanding reliability. The ACT1210E series is capable of supporting multidrop topologies, which enables multiple ECUs to be connected together. This is essential for upcoming driverless vehicles where number of electronic components and sensors will increase dramatically, and in order to optimise the performance all the ECUs need to work in sync. The ACT1210E series' characteristics are suited to

these needs and contribute to improving the quality of telecommunication and suppressing noise in an automotive system.

TDK Corporation

<https://www.tdk.com>

First software vision sensor

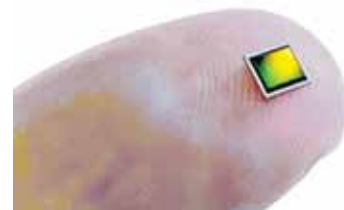
The Visual Behaviour has released Aloception, the world's first software vision sensor. Aloception is capable of automatically navigating through crowded areas by detecting obstacles using only a camera. It does the localisation and mapping of the surrounding in real time, which makes it more accurate. It focuses first on the depth and geometric understanding of the environment before the semantic aspect of things. The Aloception can be used for navigation in various applications, such as autonomous vehicles and drones.

Visualbehavior

<https://visualbehavior.ai>

Image sensor and camera module

Spectricity's S1 is the world's first truly miniaturised mass-manufacturable multispectral image sensor and camera module for mobile devices. The sensor



combines spectral and spatial features to increase the accuracy of the camera algorithms, which enable true colours through accurate and advanced white balancing. This allows it to achieve stable performance under uncontrolled and complex user environments. The